Augmentation and leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why must a checkpoint be chosen before inspecting final evaluation results?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Check whether a transformation preserves the task label. Try the recall before opening notes.

Your learning target

I can check whether a transformation preserves the task label.

I can keep related examples in one split using **source group**s.

I can compare **augmentation** fairly without inflating independent **evidence**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Check whether a transformation preserves the task label

Keep related examples in one split using **source group**s

Compare **augmentation** fairly without inflating independent **evidence**

Retrieve what you already know

> Why must a checkpoint be chosen before inspecting final evaluation results?

My starting explanation

This week's end goal

By the end, I can explain **augmentation** and **leakage** using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Augmentation and leakage case. Separate observations, given facts and assumptions before choosing an explanation.

Return to the binary image task: class 1 means the left column sum is strictly larger than the right column sum; equal sums belong to class 0. Let d be left sum minus right sum.

An **augmentation** is useful only under a stated task and transformation rule. More image files do not automatically mean more independent **evidence**.

Operation on rows [1, 0] and [0, 0]	Result rows	d	Class
Original	[1, 0] and [0, 0]	1	1
Swap top and bottom rows	[0, 0] and [1, 0]	1	1
Swap left and right columns	[0, 1] and [0, 0]	-1	0

Swapping rows preserves each column sum, so it preserves this task label for every valid image. Swapping columns changes d to -d. For nonzero d the class changes; for d = 0 it stays class 0.

Blindly copying the original label after every flip would create incorrect **training** targets.

A transformation may be valid for one task and invalid for another. A crop can remove the **evidence** needed for a label.

Choose allowed transformations using the task definition and **training evidence**, document the assumptions, and evaluate the choice on validation data.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Augmentation and leakage. Connect each labelled input, deciding step and result.

This code assumes valid 2 by 2 binary lists. It constructs new lists instead of changing the **input**. The label function encodes the classroom task rule, not a general method for recovering unknown real-world labels.

```
def difference(x):
    return x[0][0] + x[1][0] - x[0][1] - x[1][1]

def label(x):
    return int(difference(x) > 0)

def swap_rows(x):
    return [x[1][:], x[0][:]]

def swap_columns(x):
    return [row[::-1] for row in x]

x = [[1, 0], [0, 0]]
assert label(x) == label(swap_rows(x)) == 1
assert label(swap_columns(x)) == 0
tie = [[1, 0], [0, 1]]
assert label(tie) == label(swap_columns(tie)) == 0
assert x == [[1, 0], [0, 0]]
```

If an allowed transformation changes a known target, the transformed example needs a justified new label. Do not invent labels for ambiguous cases.

Keep the source ID, transformation and resulting label with every derived image so a teacher can audit what happened.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Augmentation and leakage. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Move D's flipped copy into **training**.

Predicted result

Prediction: Final-test group separation fails.

Why the result follows

Reason: The transformed D image is still derived from the held-out source, despite a new filename.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Augmentation and leakage to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Augmentation	Creating transformed training examples under a declared procedure
Label preservation	A transformation leaving the correct target unchanged for the stated task
Source group	An identifier joining records that must stay together when splitting
Leakage	Information crossing an evaluation boundary in a way that undermines the intended independent check

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Augmentation and leakage, predict a step, then compare with its explanation.

Every variant stays with its source group

SOURCE GROUP	ASSIGNED SPLIT	DERIVED RECORDS
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

A training image has source ID A. Its row-swapped copy is called A2 and placed in validation. No filenames match across splits. Is the split valid for evaluating unseen source images? Describe the repair.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Augmentation and leakage

Keep every augmented image with its original **source group**.

The facts we will use

Supplied facts

Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/24

Augmentation and leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Track provenance

Each flipped copy retains its source ID and parent image reference.





2. Assign by source

A/B originals and flips give4 training records; C gives 2 validation; D gives 2 final.





3. Check label meaning

Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

CHANGE ONE THING

Move D's flipped copy into training.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Track provenance	Each flipped copy retains its source ID and parent image reference.
2. Assign by source	A/B originals and flips give4 training records; C gives 2 validation; D gives 2 final.
3. Check label meaning	Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

What this example does not establish

Keep the stated limit

Augmentation validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

Keep the stated limit

Read the labelled picture: How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Move D's flipped copy into **training**. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Augmentation and leakage claim and explain a repair using the supplied evidence.

Check the claim against the facts

Week 23 used deliberately constructed toy images. Its validation E is exactly a row-swapped version of **training** A, and F is a row-swapped version of C.

They had different pixel arrays, but the transformation relation means they are a weak independence test for an experiment that trains on row-swapped variants.

If those variants are added, they become exact **training**-validation duplicates. Preserve the original narrow claim and design a new grouped split for the **augmentation** experiment.

Check the claim against the facts

Source groups should match the evaluation question. If the goal is generalisation to new objects or people, several photographs of the same object or person may need one group even when they have different original file IDs.

A lineage audit catches recorded relationships; an exact-pixel audit catches duplicates with inconsistent IDs. Neither audit alone proves that all dependence has been removed.

Check the claim against the facts

When a leak is discovered after validation has influenced choices, repair the split and rerun **training** and selection.

Do not merely rename or remove the overlapping files and keep claiming the old score evaluates the repaired experiment. Final data that influenced decisions is no longer reserved.

Locate the earliest unsupported step

Training contains P original and Q original. Validation contains P crop and R original. Final test contains S original. P crop is derived from P original.

Identify the overlap and explain what must be rerun if this validation set already selected a model.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Augmentation and leakage with support, then explain the deciding step in your own words.

For rows [0, 1], [0, 1], compute d and class before and after swapping rows and after swapping columns.

A training image has source ID A. Its row-swapped copy is called A2 and placed in validation. No filenames match across splits. Is the split valid for evaluating unseen source images? Describe the repair.

Explain your prediction to a partner before comparing results.

Practise with less help: Keep the equality boundary

Use workbook W2 for the connected example "**Augmentation** and **leakage**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Augmentation** and **leakage**. Proposed explanation: "Every image flip preserves the label". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

Run the investigation

> Investigate Augmentation and leakage. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

> For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

```
def difference(x):
    return x[0][0] + x[1][0] - x[0][1] - x[1][1]

def label(x):
    return int(difference(x) > 0)

def swap_rows(x):
    return [x[1][:], x[0][:]]

def swap_columns(x):
    return [row[::-1] for row in x]

x = [[1, 0], [0, 0]]
assert label(x) == label(swap_rows(x)) == 1
assert label(swap_columns(x)) == 0
tie = [[1, 0], [0, 1]]
assert label(tie) == label(swap_columns(tie)) == 0
assert x == [[1, 0], [0, 0]]
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	

Record	Your evidence
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Augmentation and leakage. Explain why each step is justified.

Explain → apply → challenge

For rows [0, 1], [0, 1], compute d and class before and after swapping rows and after swapping columns.

Training contains P original and Q original. Validation contains P crop and R original. Final test contains S original. P crop is derived from P original.

Identify the overlap and explain what must be rerun if this validation set already selected a model.

Eight source images each have an original plus two variants. Compare four epochs of these 24 records with four epochs of the eight originals, using one-example batches.

Give both update counts and one way to match budgets without using final outcomes.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **augmentation** and **leakage evidence** sheet to Image-pattern Network Investigation. Show how you can check whether a transformation preserves the task label.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Suppose 12 **training** source images each produce an original plus two variants. That creates 36 records but still only 12 **source group**s. Keep validation and final groups separate.

Report source counts as well as record counts; repeated variants are correlated **observation**s, not new independent subjects.

> **An augmented run uses 36 records for five epochs while a baseline uses 12 records for five epochs. Both use one-example batches. Are their update budgets matched? Propose a controlled comparison and explain what 36 records does not establish.**

With one-example batches the runs use 180 and 60 updates. A declared comparison could give both 180 updates, cycle the baseline data and record the exposure policy.

Keep starting parameters, architecture and evaluation cases matched where possible. Select **augmentation** settings using validation **evidence** and report the limits of the chosen budget.

Next week moves from images to sequence representations.

Eight source images each have an original plus two variants. Compare four epochs of these 24 records with four epochs of the eight originals, using one-example batches.

Give both update counts and one way to match budgets without using final outcomes.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Augmentation and leakage case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Every variant stays with its source group

SOURCE GROUP	ASSIGNED SPLIT	DERIVED RECORDS
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Design a grouped split independently

Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Augmentation and leakage, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Keep every augmented image with its original **source group**.

By the end, I can	How I will show it
Check whether a transformation preserves the task label	Explain the deciding step, show a trace or test, and state one limit.
Keep related examples in one split using source groups	Explain the deciding step, show a trace or test, and state one limit.
Compare augmentation fairly without inflating independent evidence	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Augmentation and leakage without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Check whether a transformation preserves the task label

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep every augmented image with its original **source group**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

1. Assign by source

2. Track provenance

3. Check label meaning

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check label meaning Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move D's flipped copy into **training**. Which conclusion is supported by the supplied facts?

1. Final-test group separation fails.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Final-test group separation fails. The transformed D image is still derived from the held-out source, despite a new filename.

Repair a missing building block

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Debugging shapes and gradients (ai-5/18). Goal: Check dimension contracts before computing a layer. Short practice: What should a debugging record pair with every gradient value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

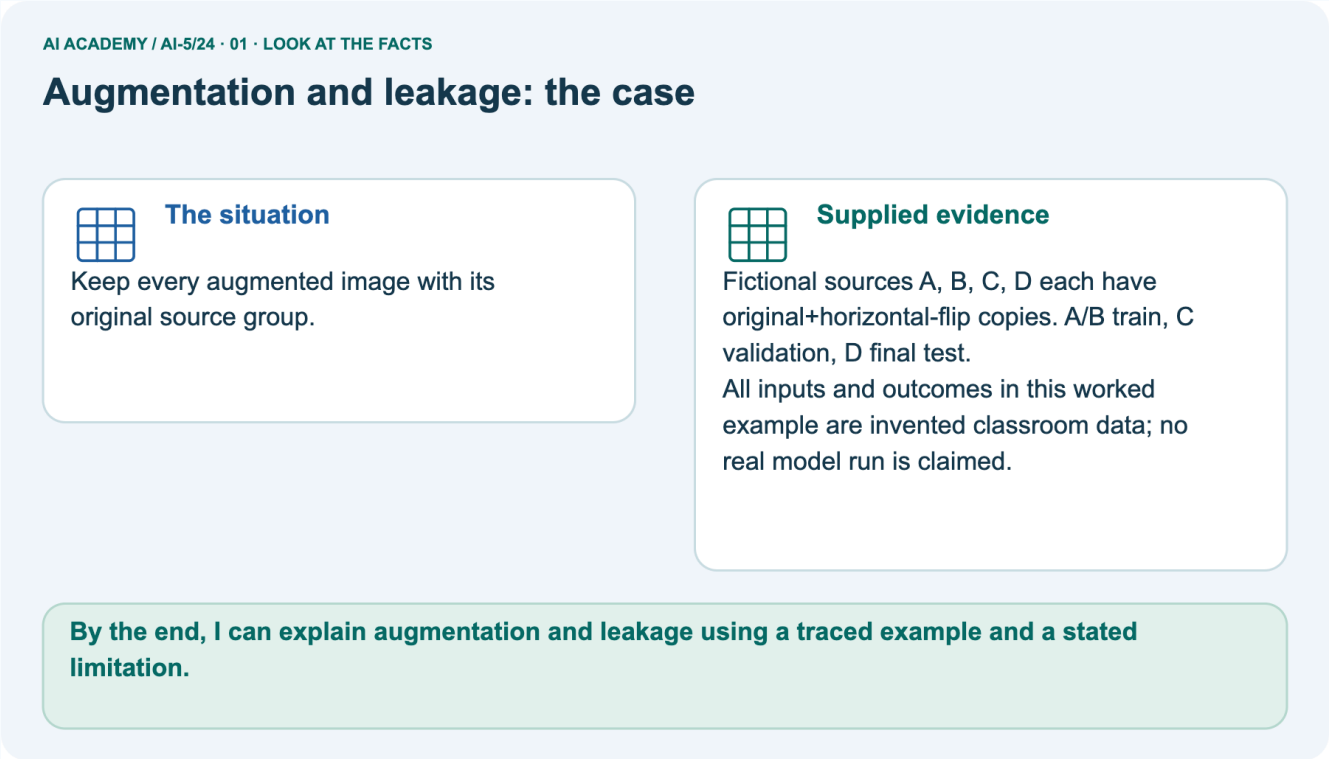
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Augmentation and leakage graphic using its labels, deciding evidence and relevant limitation.

01. Augmentation and leakage: the case



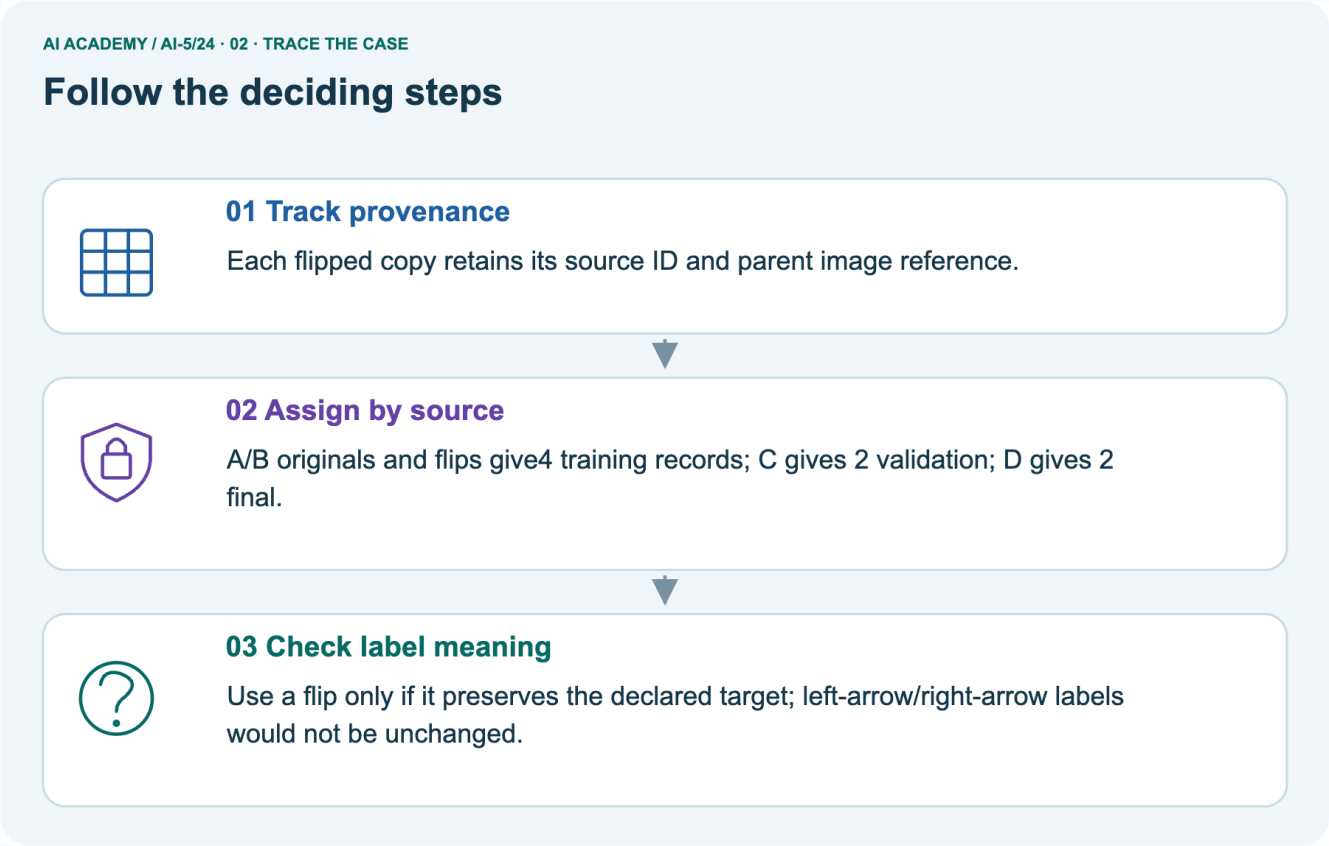
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep every augmented image with its original **source group**. Supplied facts: Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Track provenance: Each flipped copy retains its source ID and parent image reference. → Assign by source: A/B originals and flips give4 **training** records; C gives 2 validation; D gives 2 final. → Check label meaning: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Every variant stays with its source group

AI ACADEMY / AI-5/24 · 03 · INSPECT THE LESSON MODEL

Every variant stays with its source group

Source group	Assigned split	Derived records
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

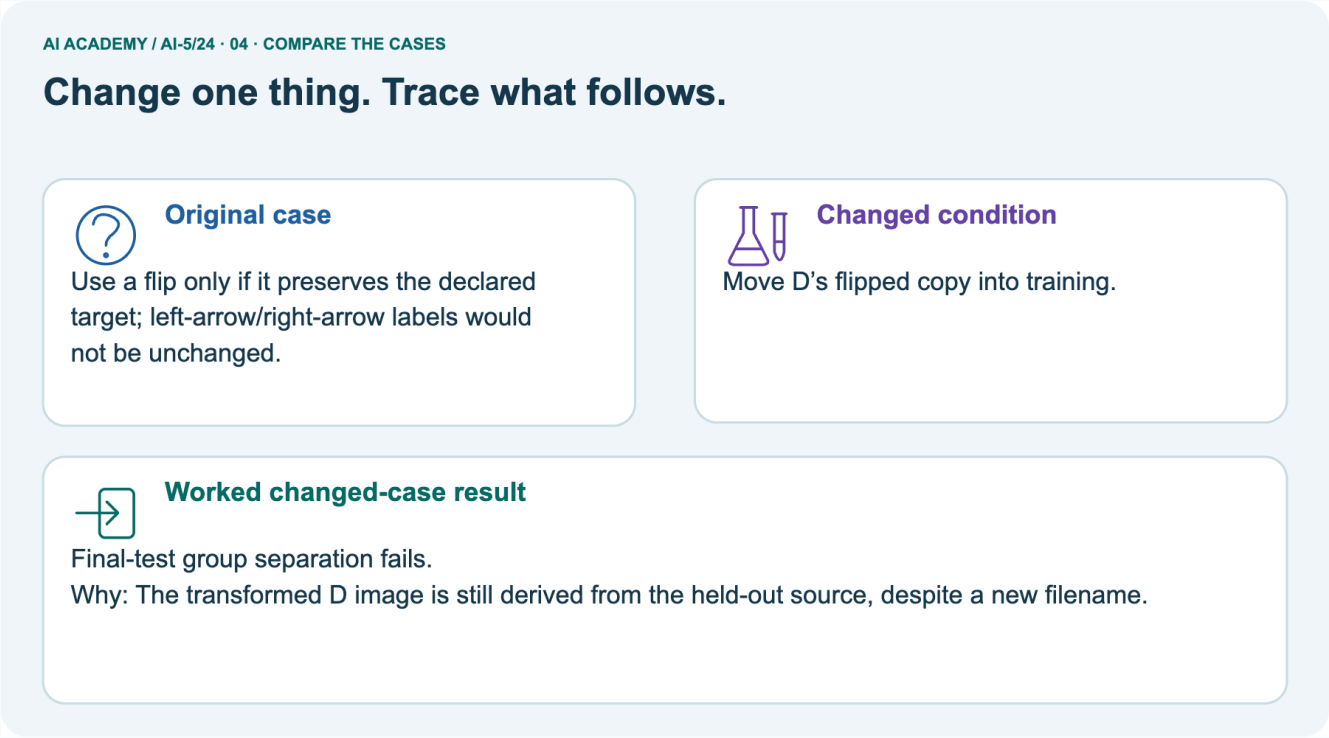
Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

Follow the labelled picture with the matching explanation. Open the vector diagram

Every variant stays with its **source group**. A | **Training** | A original and A row swap. B | Validation | B original only. C | Final test | C original only

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



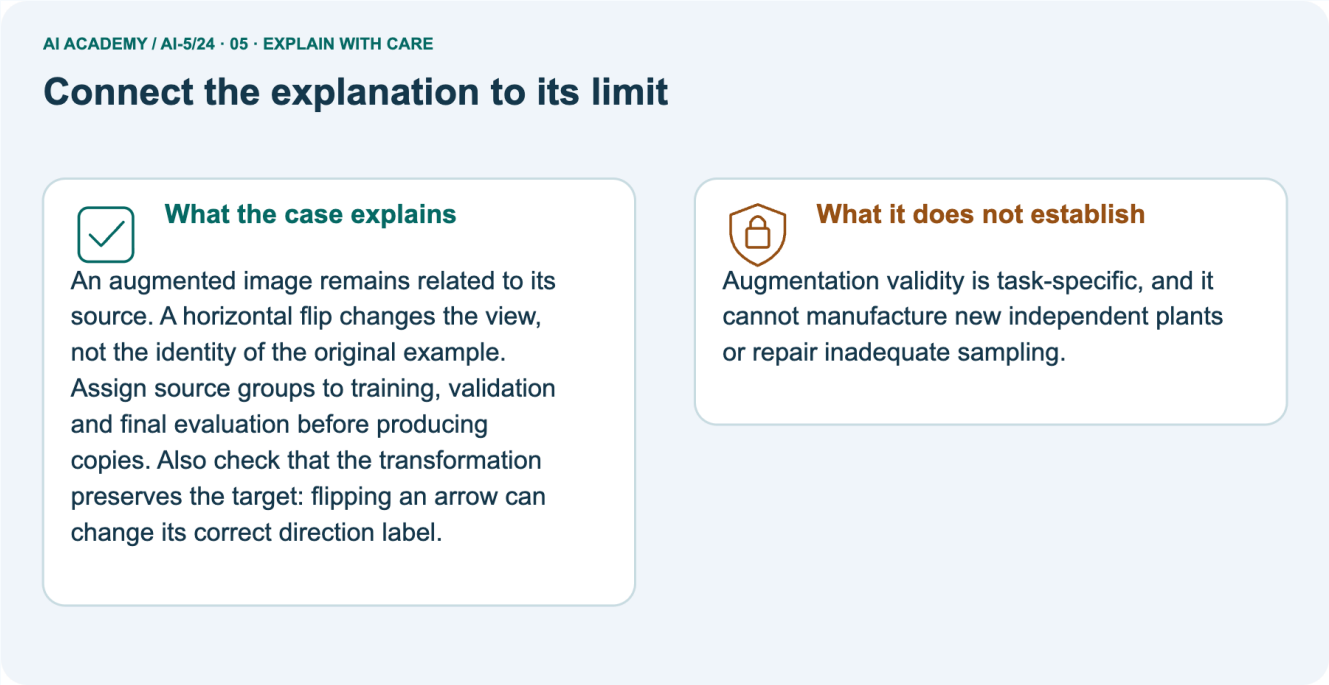
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged. Changed condition: Move D's flipped copy into **training**.

Expected result: Final-test group separation fails. Reason: The transformed D image is still derived from the held-out source, despite a new filename.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

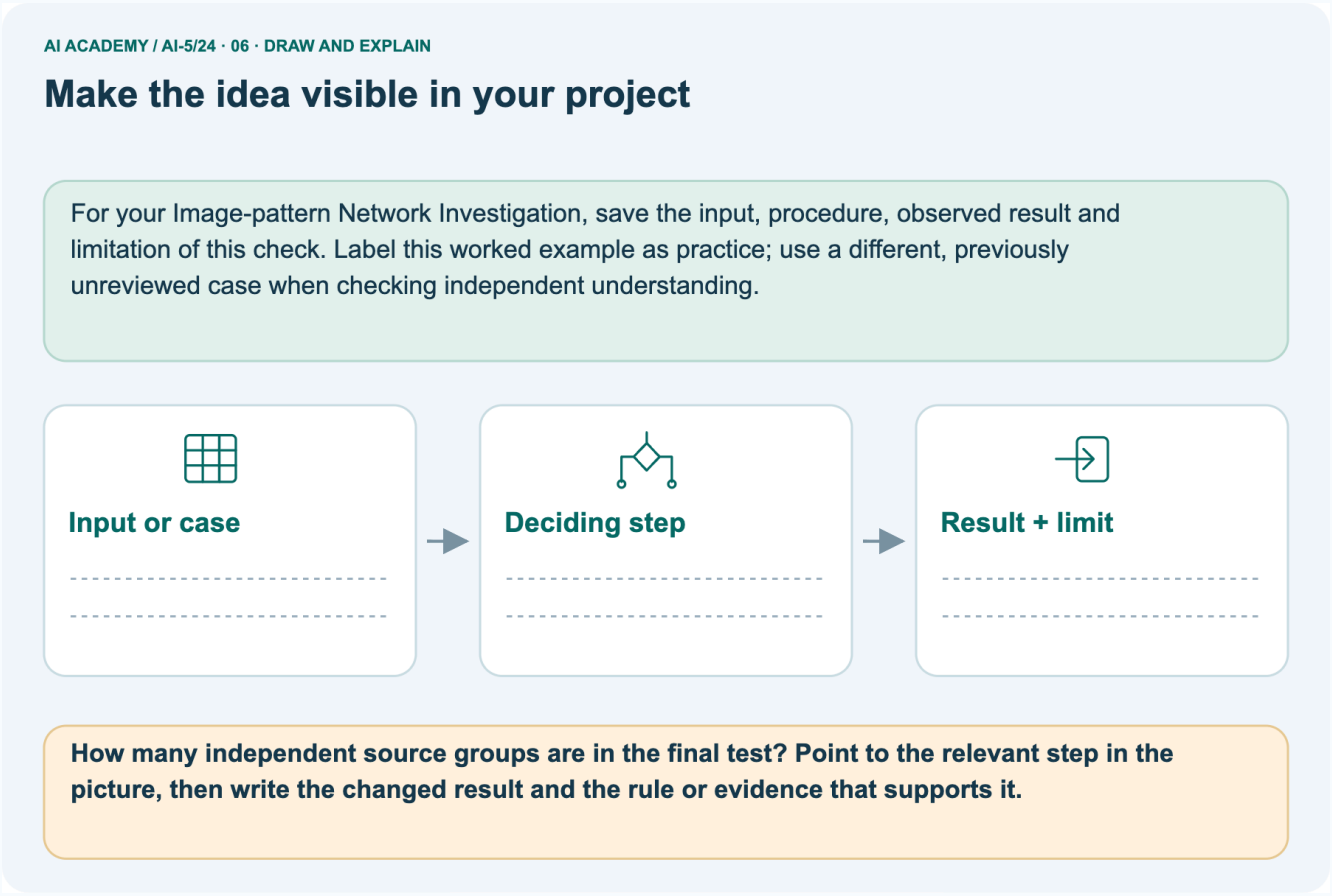
An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Limit: **Augmentation** validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.